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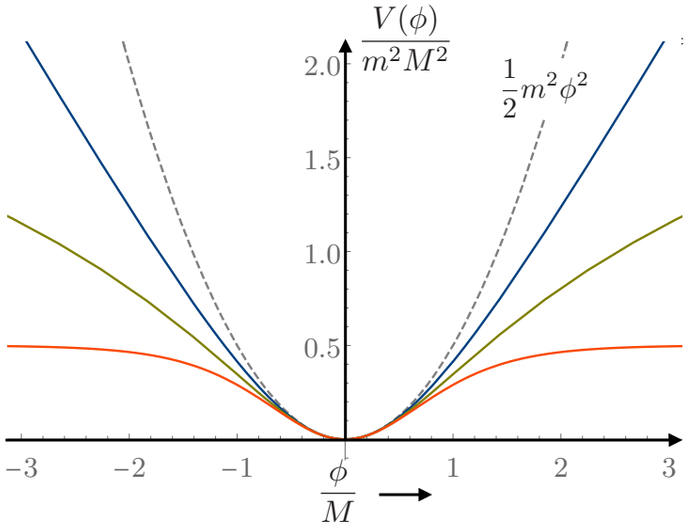
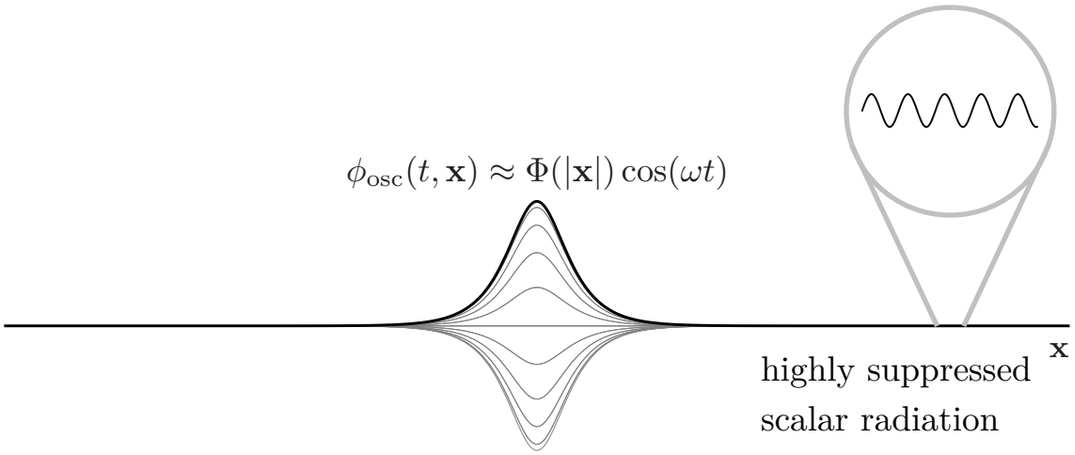
Classical Decay Rates of Oscillons

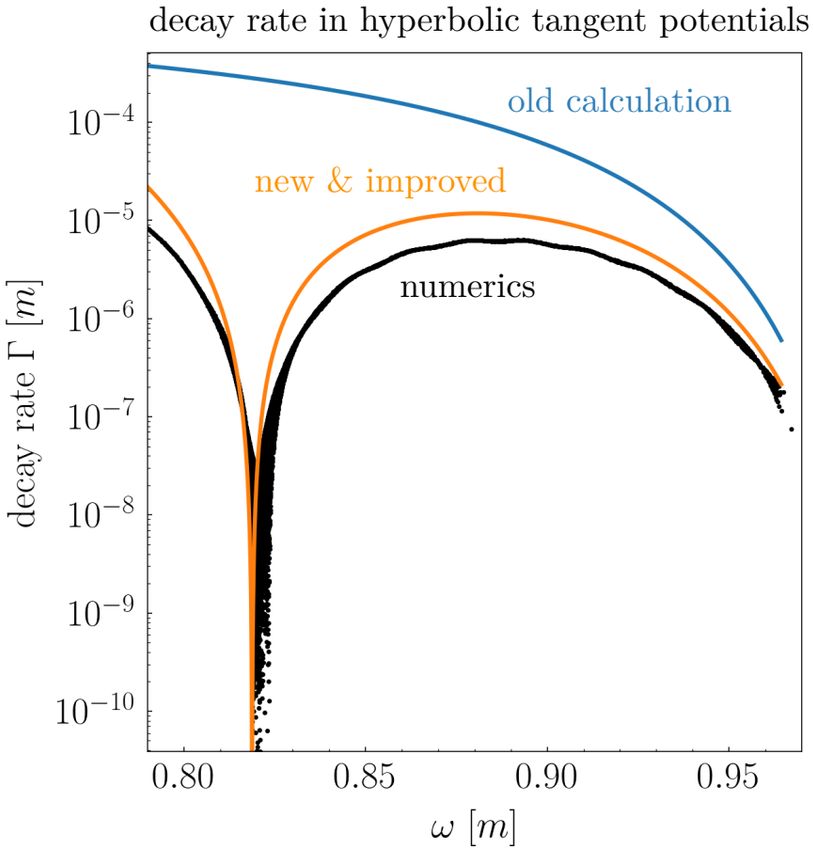
arXiv: 2004.01202





Oscillons are very long-lived, spatially localized, field configurations whose profile oscillates in time. They may exist in the inflaton field, dark-matter fields, etc.

Oscillons can survive for millions to even billions of their own oscillations (classically). Why are they so stable?

We calculate and explain the exceptionally small decay rates of oscillons (including large amplitude oscillons). In some cases, we find that decay rates can be non-monotonic in oscillon parameters, with the *dominant* radiation mode absent for certain configurations.

The key improvement in our calculation compared to earlier works comes from systematically including a spacetime-dependent effective mass term in the radiation calculation.